Ahhyun Lucy Lee

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Education

Seoul National University Sep '24 – Current
Integrated M.S. and Ph.D. Program in Brain and Cognitive Sciences
Seoul, Korea

Princeton University Apr '26 – Sep '26
Visiting Student Research Collaborator, Department of Psychology
Princeton, NJ, USA

Pohang University of Science and Technology (POSTECH) Mar '19 – Aug '24
Bachelor of Science in Life Science, Minor in Computer Science and Engineering
Pohang, Korea — incl. exchange semester at UIUC, Department of Psychology (Spring 2023)

Publications

† First author * Corresponding author

BAT-CLIP: Trimodal Alignment of Brain, Audio and Text *IEEE MLSP 2026* (Oral)
Suhyun Kim†, Jinmo Han†, Danny Dongyeop Han†, Ahhyun Lucy Lee, Jewoon Lee, Yonghyeon Gwon, Zach Paris, Chun Kee Chung, Saewoong Bahk, Nam Soo Kim, Seong Jae Hwang, Jiook Cha*

DIVER-0: A Fully Channel Equivariant EEG Foundation Model *ICML 2025 Workshop on GenBio* (Spotlight)
Danny Dongyeop Han†, Ahhyun Lucy Lee†, Taeyang Lee†, Yonghyeon Gwon†, Sebin Lee†, Seongjin Lee†, David Keetae Park, Shinjae Yoo*, Jiook Cha*, Chun Kee Chung*
arXiv:2507.14141

Conference Presentations

NeuroBabyLM: A Benchmark for Brain Alignment in Language Models Trained on Developmentally-Plausible Datasets *CCN 2026* (Poster)
Ahhyun Lucy Lee†, Danny Dongyeop Han†, Nikunj Agarwal, Kijung Son, Jiook Cha*
Poster

Enhancing psychiatric disorder diagnosis: integrating multilabel metadata with contrastive learning on T1 structural MRI *OHBM2025* (Poster)
Danny Dongyeop Han†, Ahhyun Lucy Lee†, Junbeom Kwon, Jiook Cha
Poster

The Impact of Low Motion Smoothness on Auditory Processing *ICMPC17* (Oral)
Ahhyun Lucy Lee†, Jinhee Kim, You Jeong Hong, Kyung Myun Lee*, Kyogu Lee*
Paper

Preprints & Publications in Progress

DIVER-1 : Deep Integration of Vast Electrophysiological Recordings at Scale *Under review*
Danny Dongyeop Han†, Yonghyeon Gwon†, Ahhyun Lucy Lee†, Taeyang Lee, Seongjin Lee, Jubin Choi, Sebin Lee, Jihyun Bang, Seungju Lee, David Keetae Park, Shinjae Yoo*, Chun Kee Chung*, Jiook Cha*
Paper Link

NeuroCast: Neural Decoding Benchmarks for Naturalistic Speech in Human ECoG *In progress*
Zachary Paris†, Arnab Bhattacharjee, Ahhyun Lucy Lee, Jinmo Han, Danny Dongyeop Han, Gi-

don Peeper, Shamus Sim Zi Yang, Leonard Niekerken, Zaid Zada, Paul Steven Scotti, Jiook Cha, Uri Hasson, Itamar Jalon

Few-Shot Deep Learning Reveals Brain-Based Markers of PTSD of Sexually Assaulted Adolescents *In progress*

Seoyoung Lim†, Ahhyun Lucy Lee, Ana Carolina Coelho Milani, Jiook Cha*

Research & Internship Experience

Visiting Student Research Collaborator, Princeton University (Department of Psychology) Princeton, NJ, USA
Apr '26 – Sep '26

- Collaborated with the Hasson lab on naturalistic ECoG speech decoding, contributing to the NeuroCast benchmark (in progress).

Internship in Lablup (Opensource cloud computing and AI service company) Seoul, Korea
Jan '24 – Aug '24

- Conducted exploratory and statistical analysis and trained random forest classifiers on a public clinical EEG dataset
- Presented results in plots considering reader-friendliness and communicated with co-researchers in weekly research meetings

Internship in MARG (Music and Audio Research Group, SNU) Suwon, Korea
Jul '21 – Aug '23

- Conducted independent research on "Effect of Beat Coordination of Simultaneous Dance Stimuli on Sound Perception in Audiovisual Dance," developing a real-world observation into a novel research question.

Honors and Awards

Presidential Science Scholarship, Ph.D. Track (~$17,000/year) Mar '26 – Present

- National fellowship from the Ministry of Education and the Korea Student Aid Foundation (KOSAF), awarded to graduate researchers in the natural sciences and engineering identified as having the potential to become world-leading scientists.
- One of 120 graduate scholars selected nationwide in 2026.

National Science and Engineering Undergraduate Scholarship (full tuition, ~$16,000) Mar '19 – Aug '24

- Full-ride national scholarship (KOSAF) for future scientists and engineers, maintained throughout the undergraduate program.

Genexine Research Award (~$700) Apr '21

- Awarded to top undergraduate researchers for outstanding achievement.

Leadership & Extracurricular Activities

BCSC (Brain Cognitive Science Community), President Korea
Sep '21 – Feb '23

- Planned and hosted the semestral academic conference for the community.

POSTECH, Department of Life Science, 33rd Treasurer Pohang, Korea
Mar '21 – Dec '21

- Drew up a budget and bill statement.

Teaching & Mentoring Experience

Research Mentoring, Connectome Lab (Seoul National University) Seoul, Korea
Sep '25 – Dec '25

- Mentored a premedical undergraduate with no prior ML/DL experience to a poster presentation — transdiagnostic clustering of 2,189 psychiatric patients and EEG-based cluster classification using deep models including DIVER.

SMP(Student Mentoring Program), General Life Science (LIFE101), Student Mentor Pohang, Korea
3 semesters, 2020 – 2022

- Taught General Life Science to freshmen.

Skills

Programming — Python (PyTorch, MNE, NumPy/Pandas), R, MATLAB, C/C++

Language — Korean (Native), English (Professional, TOEFL iBT 5.5/6.0)

References

Jiook Cha — Associate Professor of Psychology, Seoul National University · connectome@snu.ac.kr

Seung Tae Baek — Associate Professor of Life Sciences, POSTECH · sbaek@postech.ac.kr

Uri Hasson — Professor, Princeton Neuroscience Institute · hasson@princeton.edu